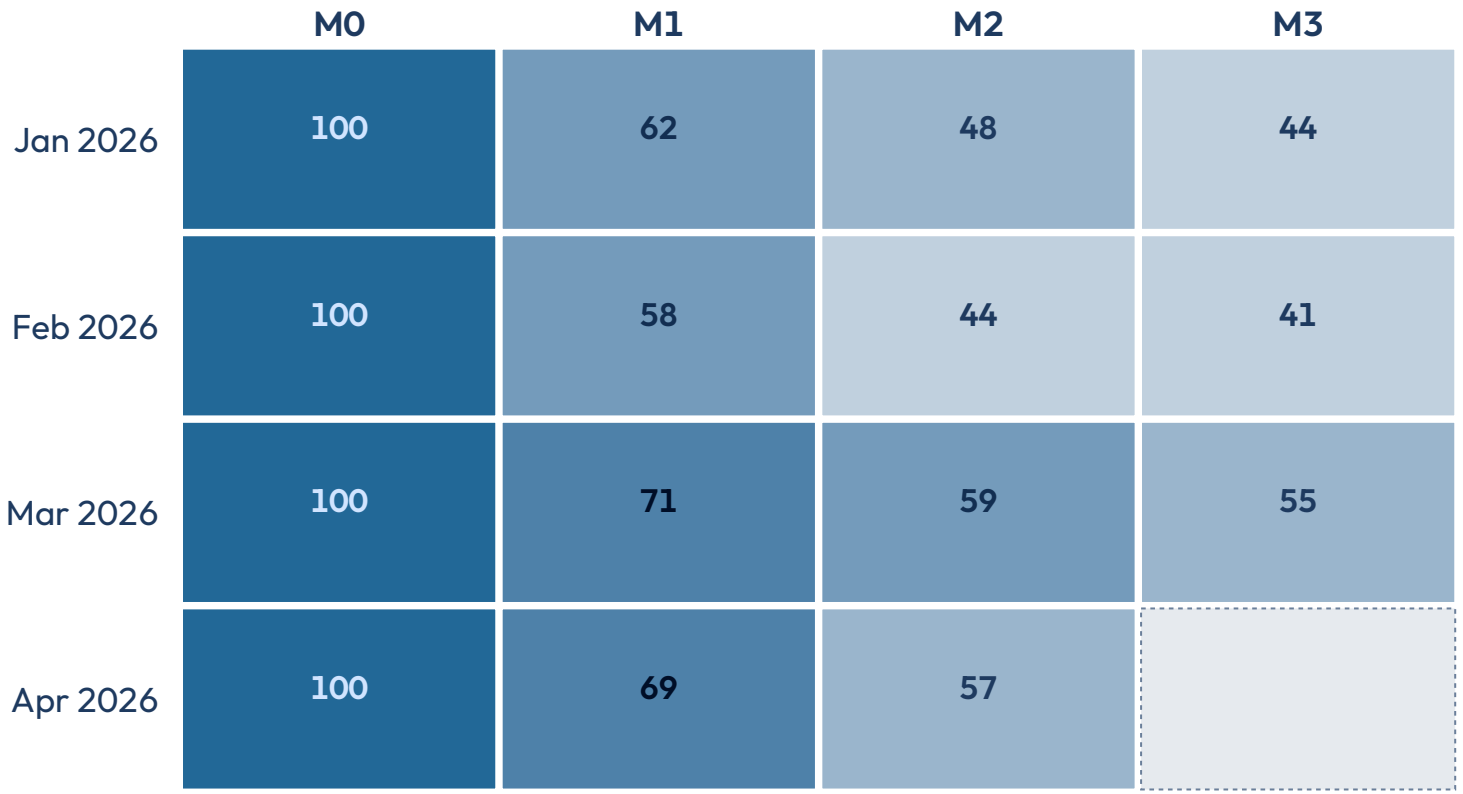
heatmap · a numeric matrix read as intensity

Where does it concentrate?

The chart family could compare magnitudes, decompose a total, plot a trend and score a 2×2 — but it had no way to put a **number at every crossing of two dimensions**. Cohort retention, a risk matrix, usage by hour and weekday: the shape a deck reaches for when the finding is a pattern rather than any single value.

Retention · 2026 cohorts

Retention decays fastest in month two.



Four cohorts, four months — the shape first, the number on it

A missing cell is not a zero.

April has no month-three number because April has not reached month three. That is the **normal** state of a cohort table, not an authoring error — so the crossing paints the bare neutral with a dotted edge and carries no intensity at all. Pad it with `0` instead and the chart states a measurement nobody took, at the bottom of the ramp, where it reads as a collapse.

The ramp is the choropleth's, not a new one.

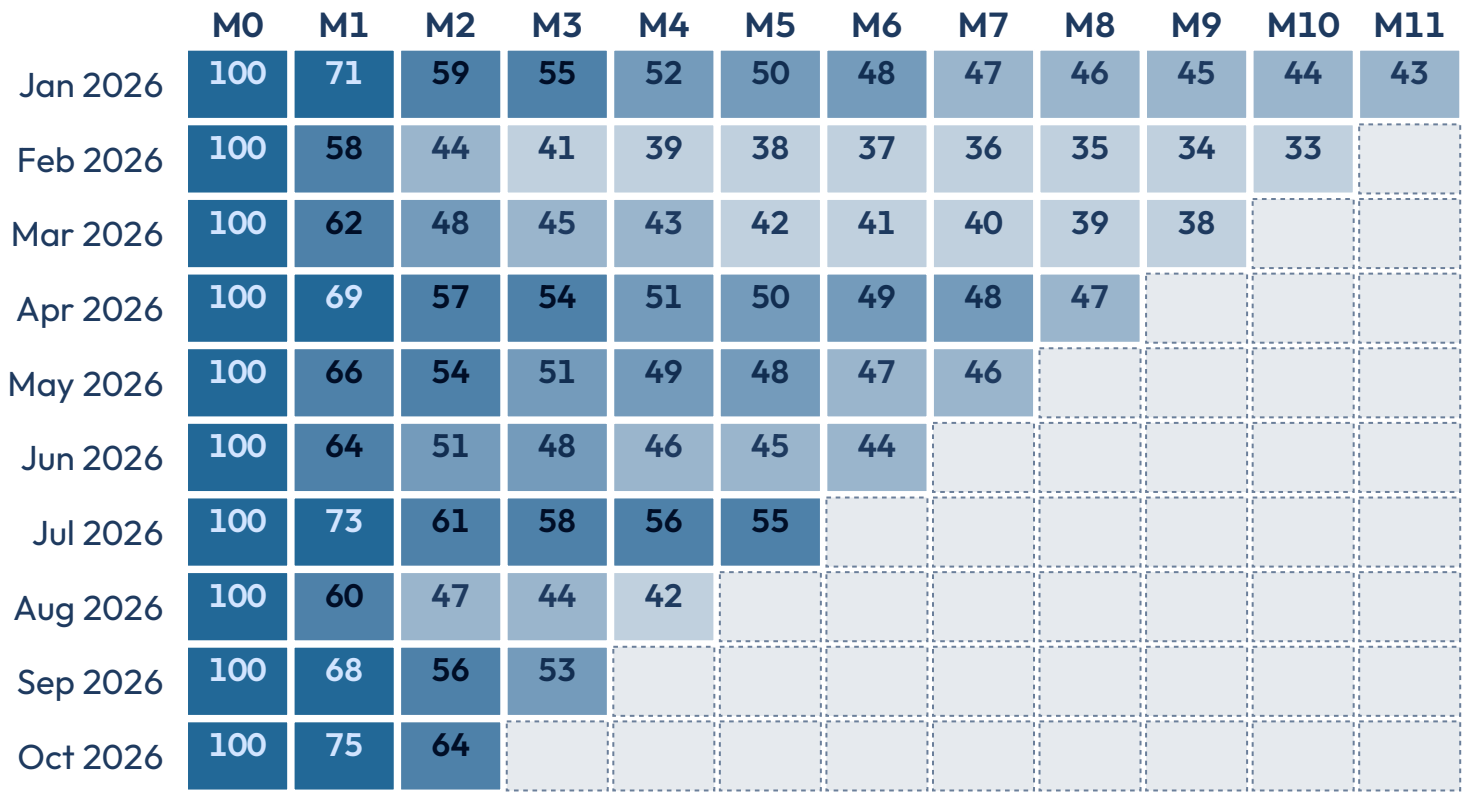
A heatmap and a `map` encode the same thing — one continuous value as one intensity — differing only in whether the cells are a grid or a geography. So this takes the mechanism `map` already proved: one hue mixed into a **neutral anchor**, never into the page. Mixing toward the canvas looks identical on light and inverts on dark, where a low value sinks *below* the empty cell it is supposed to sit above.

The ramp is five stops, so the number can sit on it.

Text on a colored fill is the one placement this family will not otherwise guarantee. The first cut printed the value on a **continuous** ramp and flipped its ink at one crossover — measured across all **33 palettes**, that leaves **318 cells** below AA, worst `2.60:1`. The fix is to stop asking for an exception: quantize. Five chosen stops means five fills to solve an ink against, and we choose which five — so the ramp steps *over* the band where no ink of any color reaches `4.5:1`. Each theme carries its own five inks, solved against the fill each stop actually paints.

Retention · the ceiling

Ten cohorts by twelve months.



When not to reach for it.

Intensity is read approximately — that is its strength for a pattern and its weakness for a comparison. One row of numbers is not a matrix; it is a `bar`, where length is judged precisely. Cells that are verbs or owners belong in `matrix-grid`. A value per country belongs on `map`. And if the decision turns on which of two similar cells is larger, the value belongs on an axis.